Singular Points

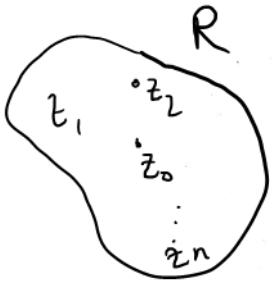
A point at which $f(z)$ fails to be analytic is called a singular point or singularity of $f(z)$.

- $f(z)$ undefined
  - ↳ $f'(z)$ does not exist
  - ↳ $f(z)$ is not differentiable

Analytic Function

If $f'(z)$ exists at all points z of a region R, then $f(z)$ is said to be analytic in R.

↳ $f'(z)$ differentiable



Various types of Singularities

- isolated singularities
- poles
- branch points

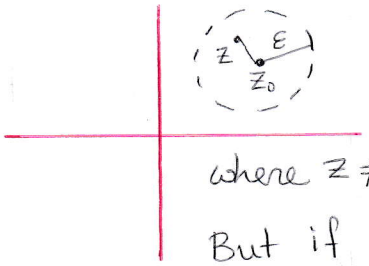
Isolated Singularities

The function f has an isolated singularity at z_0, if it is $\underbrace{\text{analytic}}$ on a deleted open neighbourhood

↳ continuous
↳ defined

of z_0, but $\underbrace{\text{not analytic}}$ at z_0.

↳ discontinuous
↳ not defined



Deleted Neighbourhood:

$0 < |z - z_0| < \varepsilon$

↳ If $f(z = z_0) =$ undefined

↓

z_0 is an isolated singularity of $f(z)$

where $z \neq z_0$

But if $z = z_0$

$\Rightarrow z - z_0 = 0$

unable to differentiate

Example

① $f(z) = \frac{1}{z}$ has isolated singularity at 0.

if $z = 0$ then $f(z) = \frac{1}{0} = \infty$

$\Rightarrow f(z)$ is undefined

$\Rightarrow f(z)$ has isolated singularity at 0.

$\frac{1}{z-0}$

$z = 0$

$f(z) = \infty$

② $f(z) = e^{\frac{1}{z-2}}$ has isolated singularity at 2.

Conside: $e^{\frac{1}{z-2}}$

if $z = 2$ then $f(z) = e^{\frac{1}{0}} = e^{\infty} = \infty$

$\Rightarrow f(z)$ undefined

$\therefore f(z)$ has isolated singularity at 2.

Poles

If n is +ve integer such that $\lim_{z \to z_0} (z-z_0)^n f(z) = A \neq 0$ } constant apart from '0' i.e. Analytic

then $z = z_0$ is called a pole of order n.

If $n=1$, then z_0 is called a simple pole

> Note: $f(z) = \dfrac{g(z)}{(z-z_0)^n}$. since $z \neq z_0$, hence we multiply $(z-z_0)^n$ with $f(z)$ as $f(z)$ contains a fraction with denominator $z - z_0$.

Examples

① $f(z) = \dfrac{1}{(z-2)^3}$

$$\lim_{z \to 2} (z-2)^3 f(z) = \lim_{z \to 2} (z-2)^3 \frac{1}{(z-2)^3}$$

$$= \lim_{z \to 2} 1$$

$$= 1 \neq 0$$

$f(z)$ has a pole of order 3 at $z = 2$

isolated singularity at $z = 2$.

② $f(z) = \dfrac{3z-2}{(z-1)^2 (z+1)(z-4)}$ isolated singularity at $z = \pm 1, 4$

$$\lim_{z \to -1} [z-(-1)]^1 f(z)$$

$$= \lim_{z \to -1} (z+1) \frac{3z-2}{(z-1)^2(z+1)(z-4)}$$

$$= \frac{-5}{4(-5)}$$

$$= \frac{1}{4}$$

$$\neq 0$$

f(z) has a pole of order 2 at $z = 1$

$$\lim_{z \to 1} (z-1)^2 f(z)$$

$$= \lim_{z \to 1} (z-1)^2 \frac{3z-2}{(z-1)^2(z+1)(z-4)}$$

$$= \frac{1}{(2)(-3)}$$

$$= -\frac{1}{6}$$

$$\neq 0$$

f(z) has simple pole at $z = -1$ (while $n = 1$)

$$\lim_{z \to 4} (z-4)^1 f(z)$$

$$= \lim_{z \to 4} (z-4) \frac{3z-2}{(z-1)^2(z+1)(z-4)}$$

$$= \frac{12-2}{(9)(5)}$$

$$= \frac{10}{45} = \frac{2}{9} \neq 0$$

f(z) has simple pole at $z = 4$ (while $n = 1$)

Branch Points

Branch points of multiple-valued functions are singular points. $\left\{ \begin{array}{l} \text{domain of } z \text{ such that } f'(z) \\ \text{is undefined} \end{array} \right.$

Examples:

1. $f(z) = (z-3)^{1/3}$

$$f'(z) = \frac{1}{3}(z-3)^{-2/3}$$

$$= \frac{1}{3\sqrt[3]{(z-3)^2}}$$

at $z = 3$, $f'(z)$ is undefined

$\Rightarrow f(z)$ is not analytic

$\Rightarrow f(z)$ has a branch point at $z = 3$.

2. $f(z) = \ln(z^2 + z - 2)$

$$f'(z) = \frac{(2z+1)}{z^2+z-2}$$

$f'(z)$ is undefined at

$$z^2 + z - 2 = 0$$
$$z^2 + 2z - z - 2 = 0$$
$$(z+2)(z-1) = 0$$
$$z = -2, 1$$

$f(z)$ has branch points at $z = 1$, $z = -2$

(Exercise Sheet 5)

8. For each of the following functions locate and name the singularities in the finite z-plane and determine whether they are isolated singularities or not.

(b) $f(z) = \dfrac{\ln(z-2)}{(z^2+2z+4)^4}$

- $f(z)$ has isolated singularity at $z \leq 2$ $\left\{ \because \ln(z-2) = \ln(2-2) = \ln 0 = \infty \right.$ or $z \leq 2 \to f(z) = \infty$

- At $z^2+2z+4=0$, $f(z)$ is undefined.
  $\therefore$ $f(z)$ has isolated singularity at $z = -1 \pm i\sqrt{3}$

- At $z^2+2z+4=0$

$$z = \frac{-2 \pm \sqrt{4-4(1)(4)}}{2}$$

$$= \frac{-2 \pm \sqrt{-12}}{2}$$

$$= \frac{-2 \pm i2\sqrt{3}}{2}$$

$$= -1 \pm i\sqrt{3}$$

$z = -1 \pm i\sqrt{3}$ are poles of order 4.

$$\frac{\ln(z-2)}{(z^2+2z+4)^4}$$

$$= \frac{\ln(z-2)}{\{z-(-1+i\sqrt{3})\}^4 \{z-(-1-i\sqrt{3})\}^4}$$

- $$f'(z) = \frac{\dfrac{(z^2+2z+4)^4}{z-2} - \ln(z-2)4(z^2+2z+4)^3(2z+2)}{(z^2+2z+4)^8}$$

$$= \frac{\dfrac{z^2+2z+4}{z-2} - 4(2z+2)\ln(z-2)}{(z^2+2z+4)^5}$$

$$= \frac{(z^2+2z+4) - 4(z-2)(2z+2)\ln(z-2)}{(z-2)(z^2+2z+4)^5}$$

But, if $z \leq 2$ then $\ln(z-2)$ will become $\ln 0 = \infty$ or $\ln(-ve) = \infty$

$\therefore$ at $z=2$, $f'(z)$ is undefined

Also $f'(z)$ is undefined at $z = -1 \pm i\sqrt{3}$ $(\because z^2+2z+4 \neq 0)$

$\therefore$ $f(z)$ has branch points at $z \leq 2, -1 \pm i\sqrt{3}$.